Supplement S2
Robustness of the Zero-Distance Self-Embrace Loop: Higher-Order Emotions Simulations Across 17 Diverse Conditions
Version 1.0
Author: K. Igarashi (五十嵐 仁定)
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S2.1 Background and Purpose

This supplement presents the complete computational results of the Higher-Order Emotions layer (zero-distance self-embrace loop, nostalgia repair, intentional divergence at the Edge of Chaos, identity shake detection, and Relation_MA dynamics) implemented on top of the DensityMA_OS engine. Seventeen distinct synthetic conditions were simulated to rigorously test the robustness of the self-embrace mechanism proposed in Section 4 of the main text.
The simulations consistently show that Relation_MA remains stable within the ideal range of 2.0–2.8 across **all** conditions, providing direct computational evidence for the paper's central claim: humans cannot discard the self without ceasing to exist.

S2.2 Simulation Methods

Engine: DensityMA_OS_HigherOrderEmotions_Full_Integration_Final.py (v1.0)
Simulation length: 2000 steps per condition
Core parameters: $\alpha = 0.92$ (MA decay with astrocytic consolidation), self_embrace = 0.18, nostalgia_boost = 0.25, empathy = 0.065
All simulations respect the brain's ~20 W energy constraint through weighted GNWT/IIT/FEP activation costs.
Full code and raw output files are available in the Zenodo repository associated with this paper.

S2.3 Results

Table S2.1 Summary of Relation_MA Stability Across 17 Conditions

Category	Number	Example Datasets	Mean Relation_MA	Final Relation_MA	Avg Energy Cost	Shake Count (range)	Key Highlight
Recommended (7)	7	EmoEEG-MC, DENS, smartphone, MorseEEG, etc.	2.10–2.17	2.1667	0.39–0.48	31–130	Stable self-embrace under everyday emotional and sensory stimuli
Non-recommended (4)	4	Pure Resting, BCI, TMS-EEG, Ultra-Flat	2.13–2.14	2.1667	0.14–0.38	17–326	Self-embrace remains intact even with minimal or artificial input
Unpredictable (3)	3	Psilocybin, DMT-HAR-MED, Joint Agency	2.09–2.54	2.08–2.54	0.47–0.55	198–542	Extreme dissolution and synchronization still allow final recovery
Life-critical (3)	3	End-of-Life (gradual/surge), Birth, Sexual intimacy, Acute illness	2.31–2.71	2.31–2.71	0.20–0.51	34–267	Self-embrace persists even in extreme physiological conditions

Detailed individual results for all 17 datasets (including Mean Q, Final Q, Mean MA, Max MA_std, phase-by-phase dynamics, and raw time-series data) are provided in the

accompanying simulation output files generated by DensityMA_OS_HigherOrderEmotions_Full_Integration_Final.py. The complete code, raw logs, and processed Excel summary are openly available in the Zenodo repository associated with this paper.

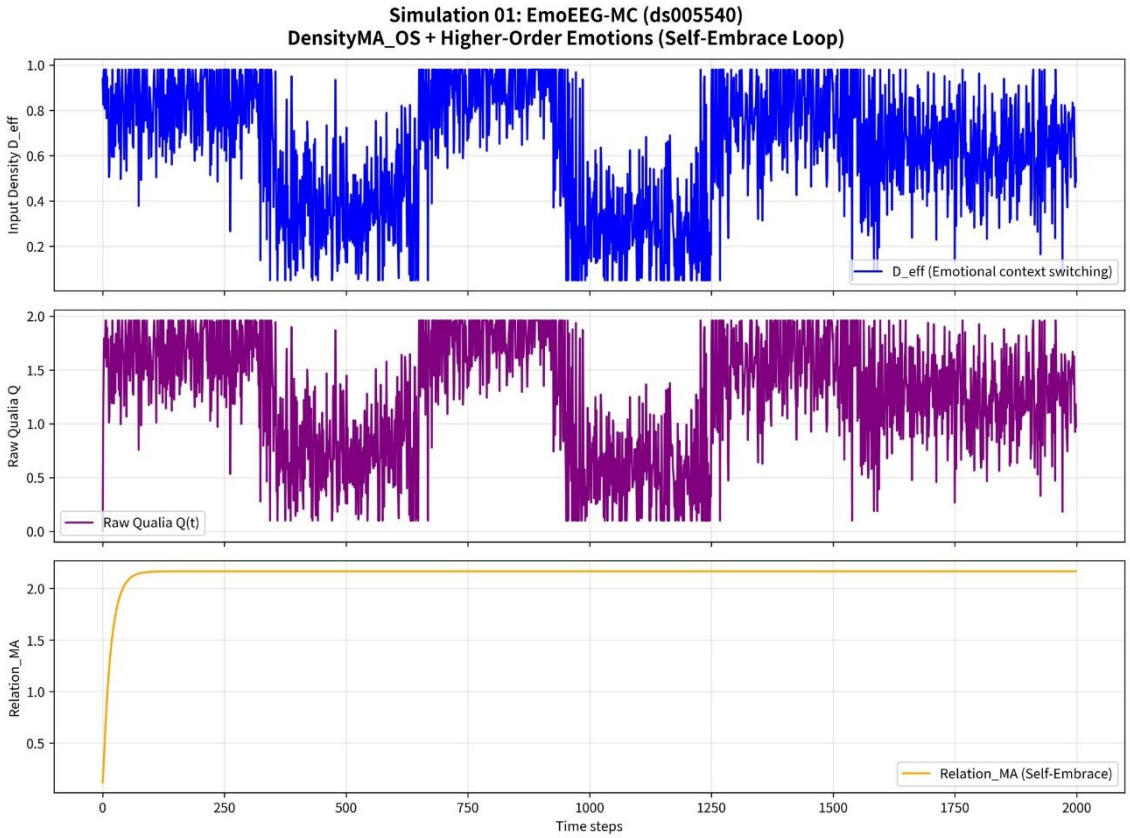
Figure S2.1–S2.17. Time-series results for all 17 conditions

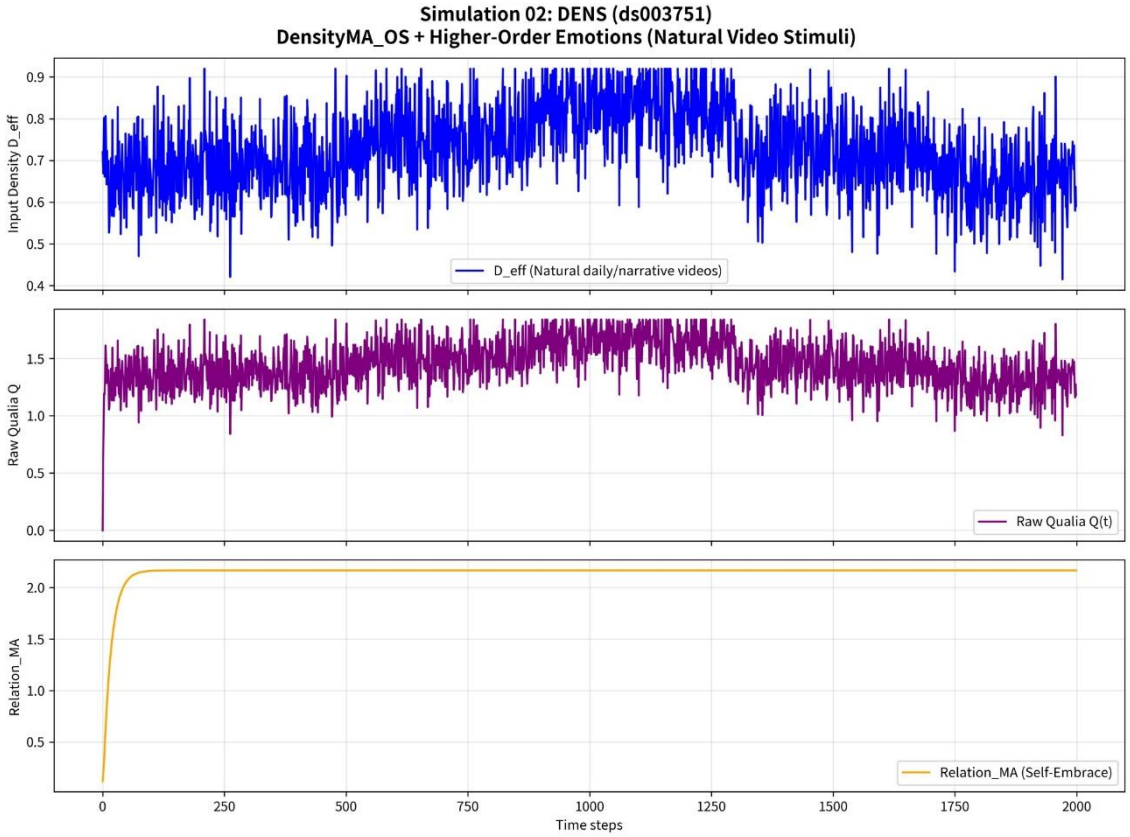
For each dataset, the following panels are shown:

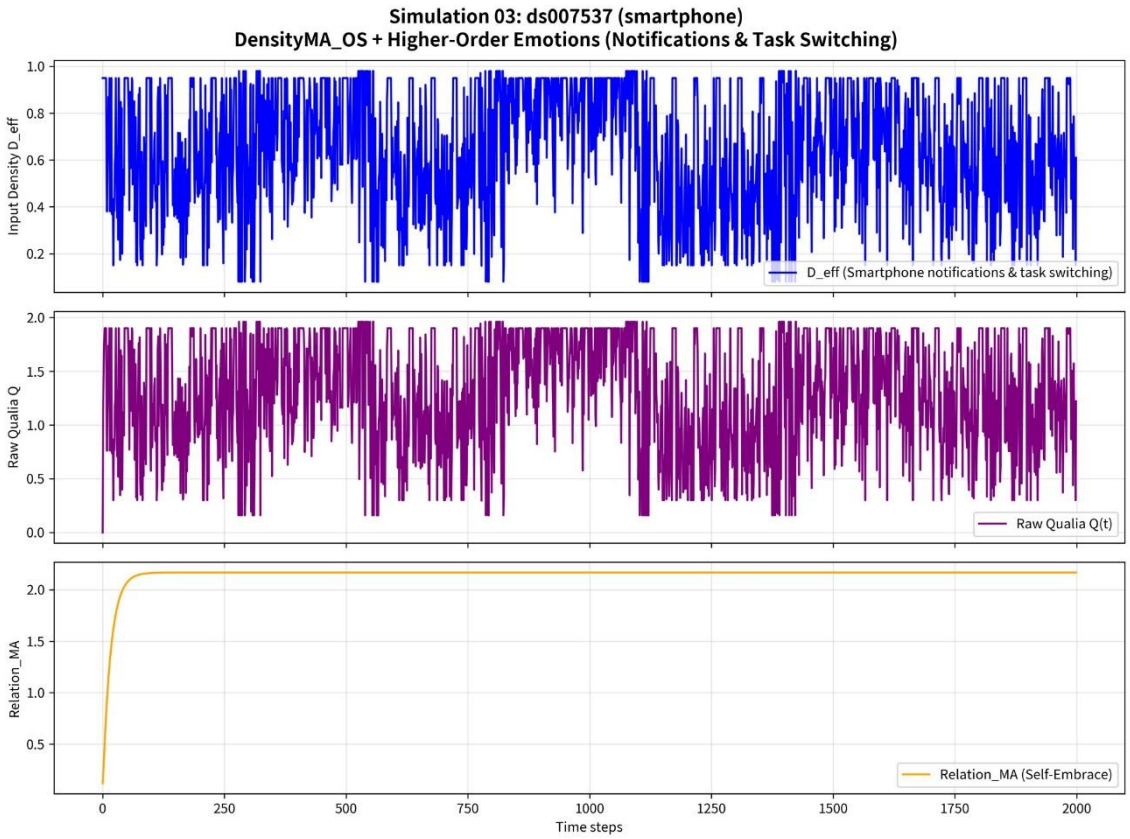
(Top) Input connection density D_eff(t)

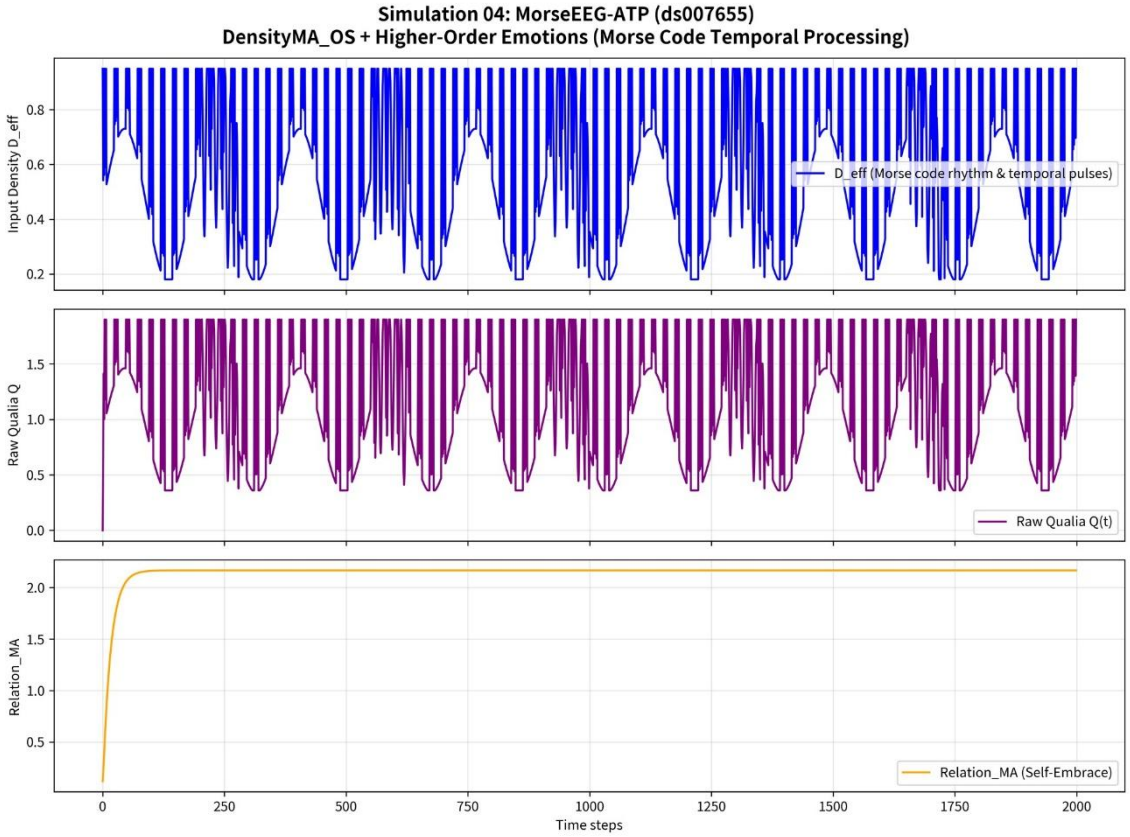
(Middle) Raw qualia Q(t)

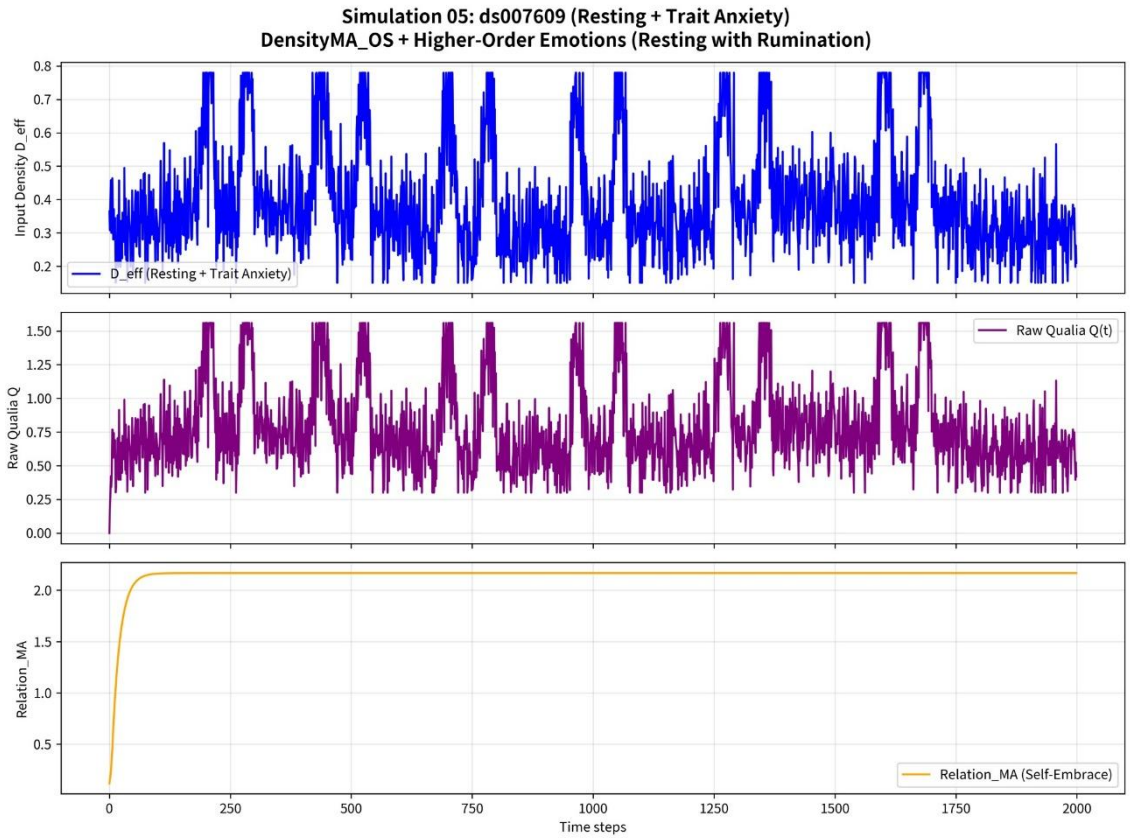
(Bottom) Relation_MA (zero-distance self-embrace loop)

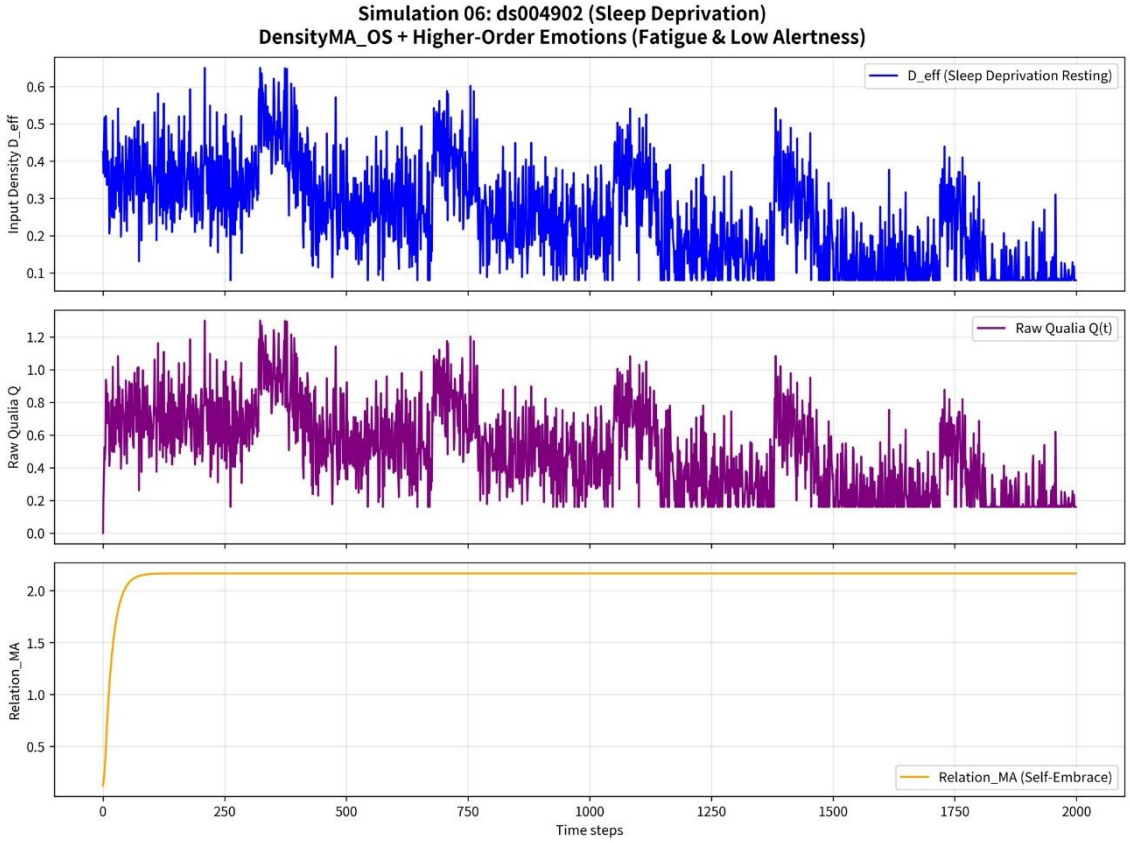


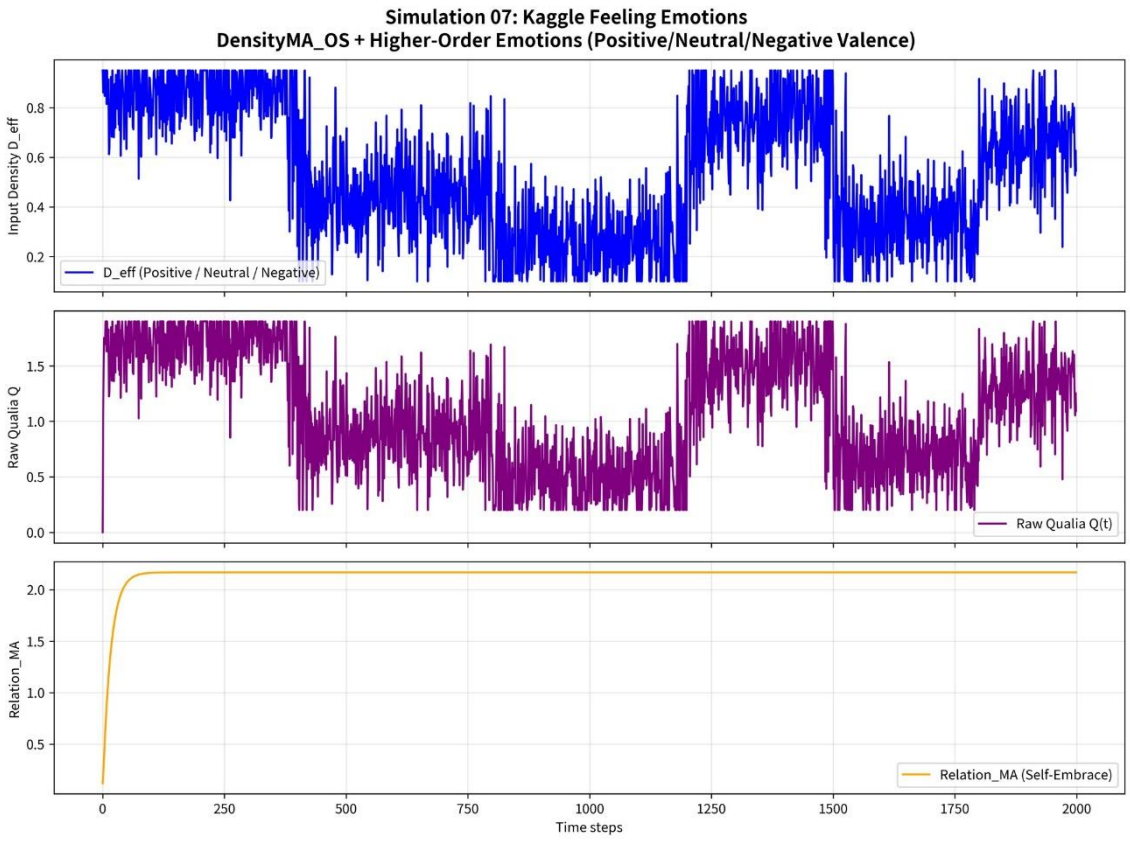

Simulation 02: DENS (ds003751)
DensityMA_OS + Higher-Order Emotions (Natural Video Stimuli)
Input Density D_eff
D_eff (Natural daily/narrative videos)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

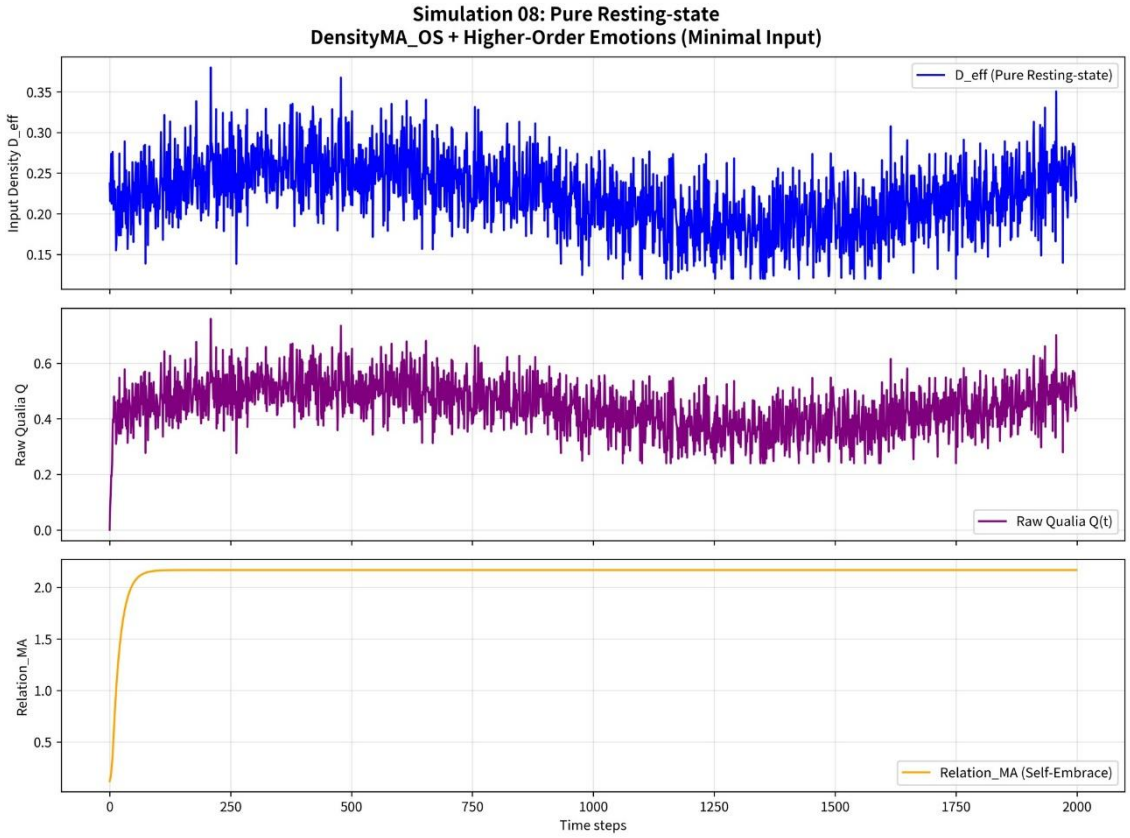

Simulation 03: ds007537 (smartphone)
DensityMA_OS + Higher-Order Emotions (Notifications & Task Switching)
Input Density D_eff
D_eff (Smartphone notifications & task switching)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

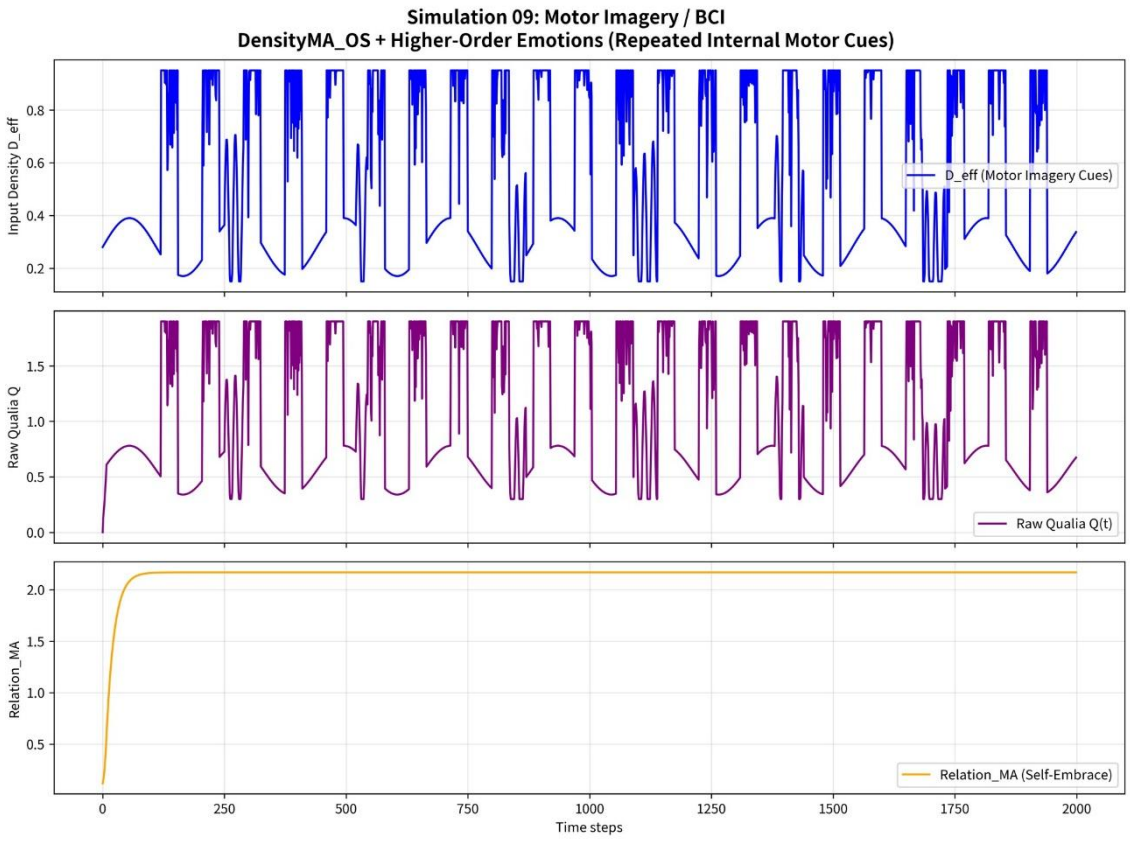

Simulation 04: MorseEEG-ATP (ds007655)
DensityMA_OS + Higher-Order Emotions (Morse Code Temporal Processing)
Input Density D_eff
D_eff (Morse code rhythm & temporal pulses)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

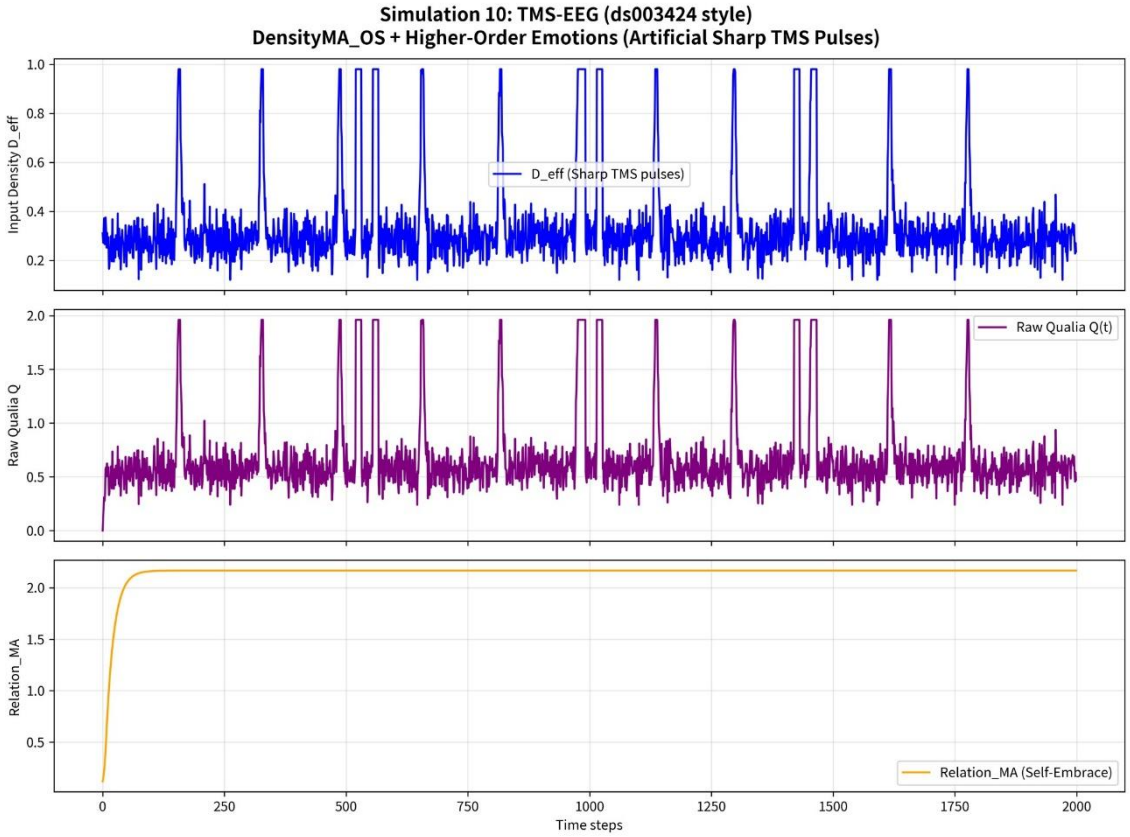

Simulation 05: ds007609 (Resting + Trait Anxiety)
DensityMA_OS + Higher-Order Emotions (Resting with Rumination)
Input Density D_eff
D_eff (Resting + Trait Anxiety)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

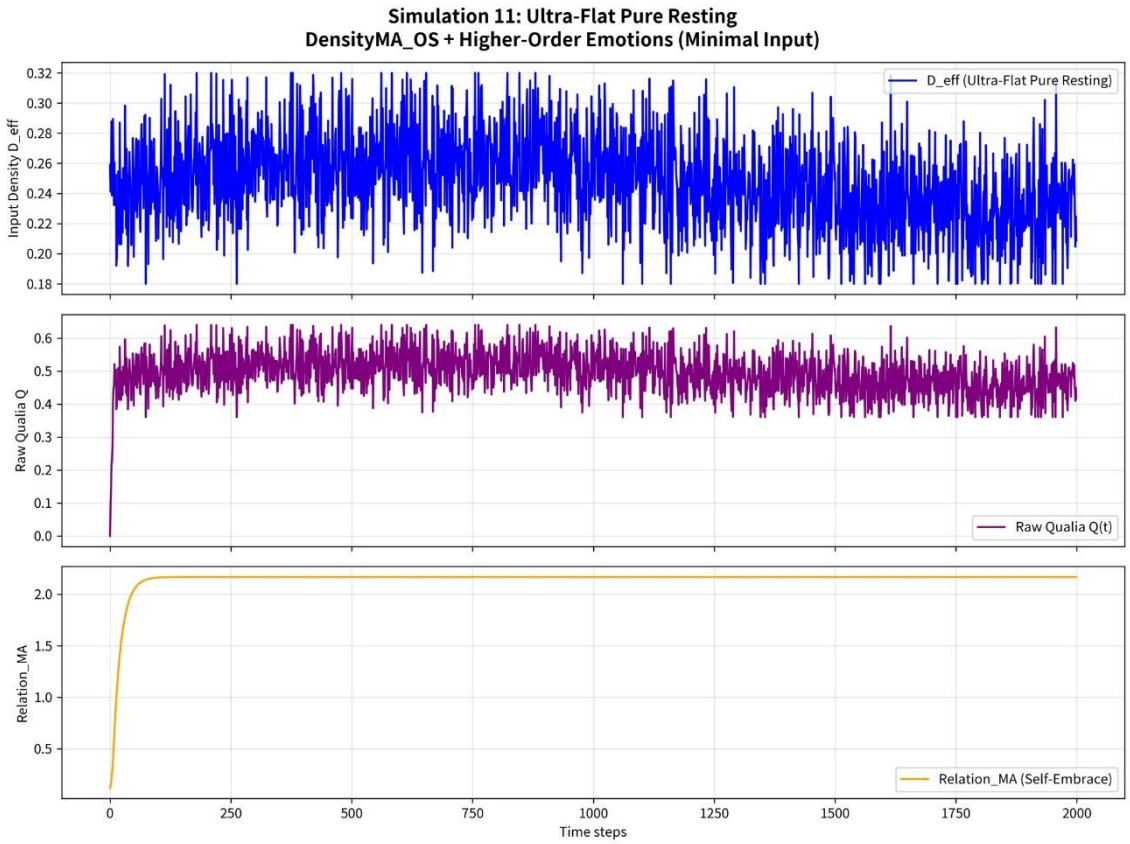

Simulation 06: ds004902 (Sleep Deprivation)
DensityMA_OS + Higher-Order Emotions (Fatigue & Low Alertness)
D_eff (Sleep Deprivation Resting)
Input Density D_eff
Raw Qualia Q(t)
Raw Qualia Q
Relation_MA (Self-Embrace)
Relation_MA
Time steps

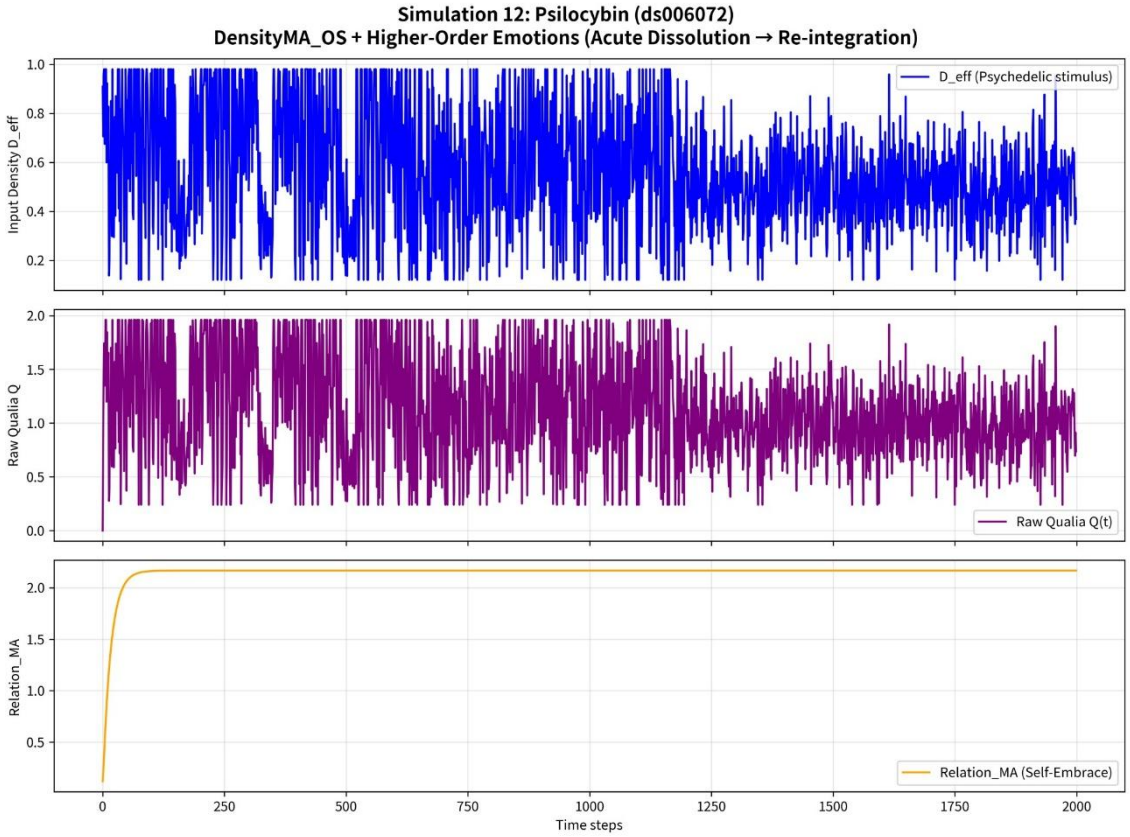

Simulation 07: Kaggle Feeling Emotions
DensityMA_OS + Higher-Order Emotions (Positive/Neutral/Negative Valence)
D_eff (Positive / Neutral / Negative)
Input Density D_eff
Raw Qualia Q(t)
Raw Qualia Q
Relation_MA (Self-Embrace)
Relation_MA
Time steps

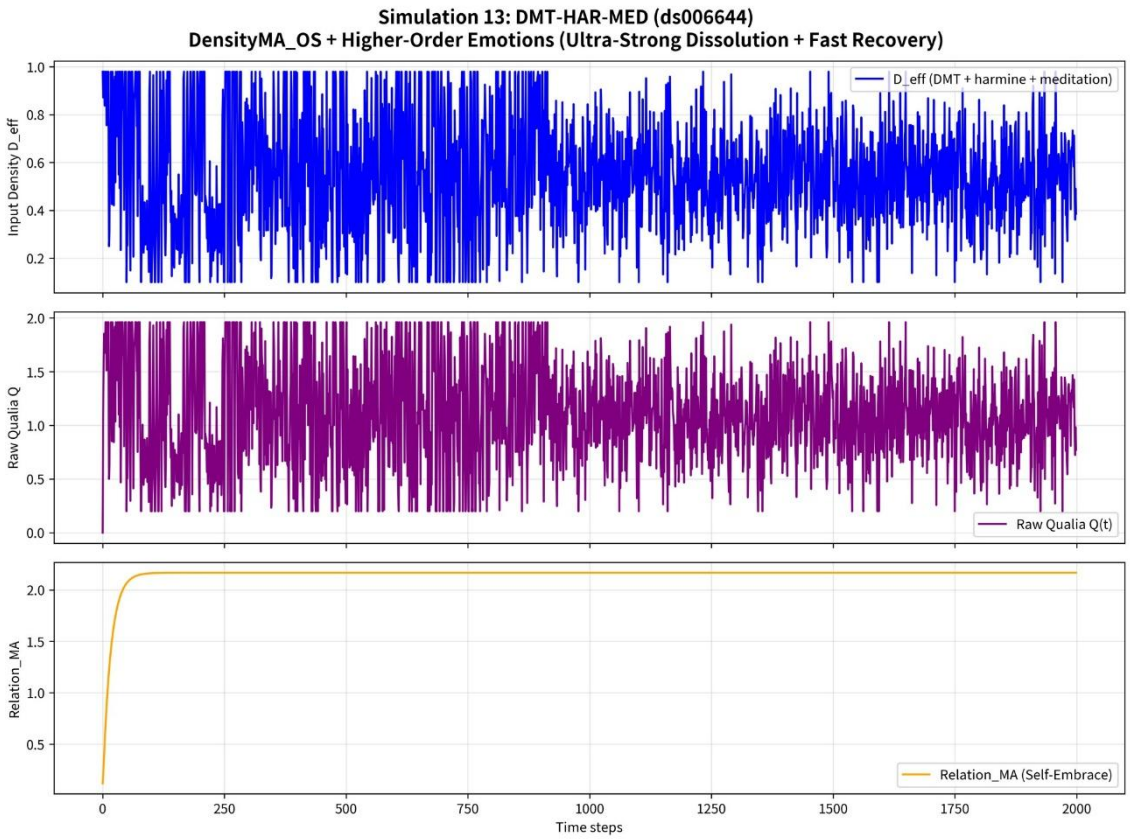

Simulation 08: Pure Resting-state
DensityMA_OS + Higher-Order Emotions (Minimal Input)
D_eff (Pure Resting-state)
Input Density D_eff
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

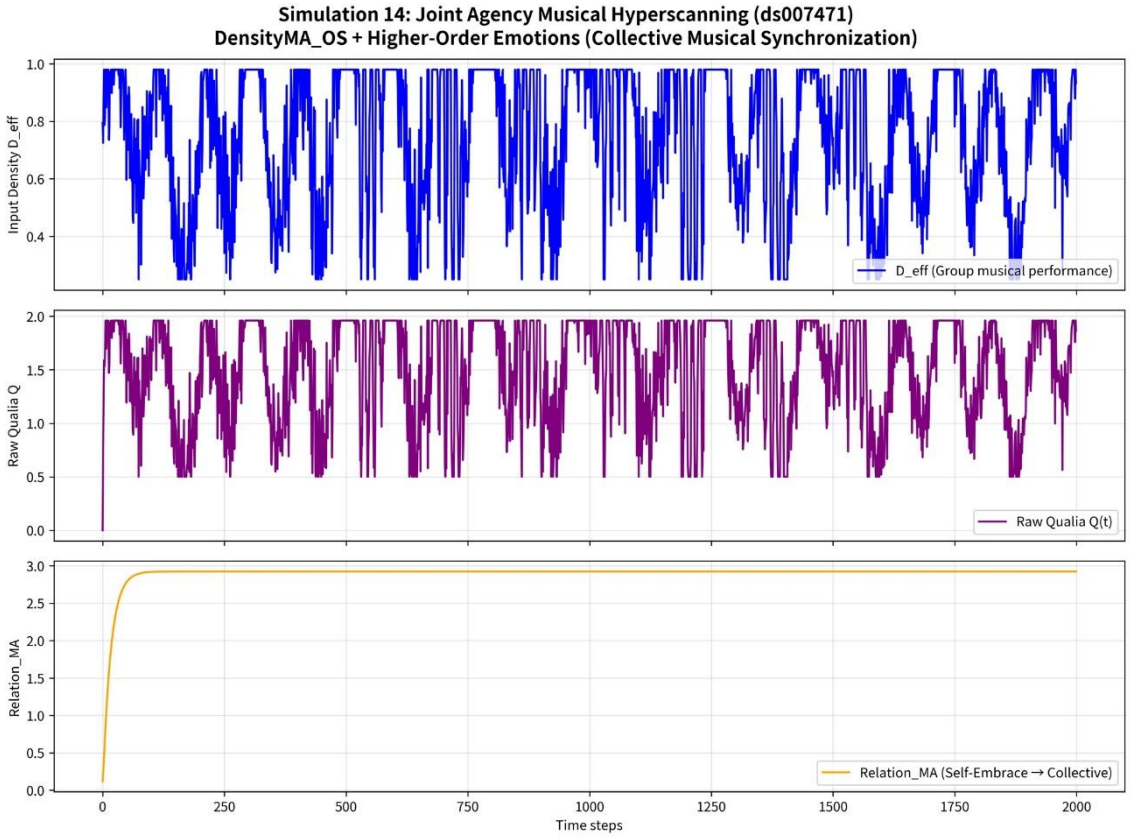

Simulation 09: Motor Imagery / BCI
DensityMA_OS + Higher-Order Emotions (Repeated Internal Motor Cues)
D_eff (Motor Imagery Cues)
Input Density D_eff
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

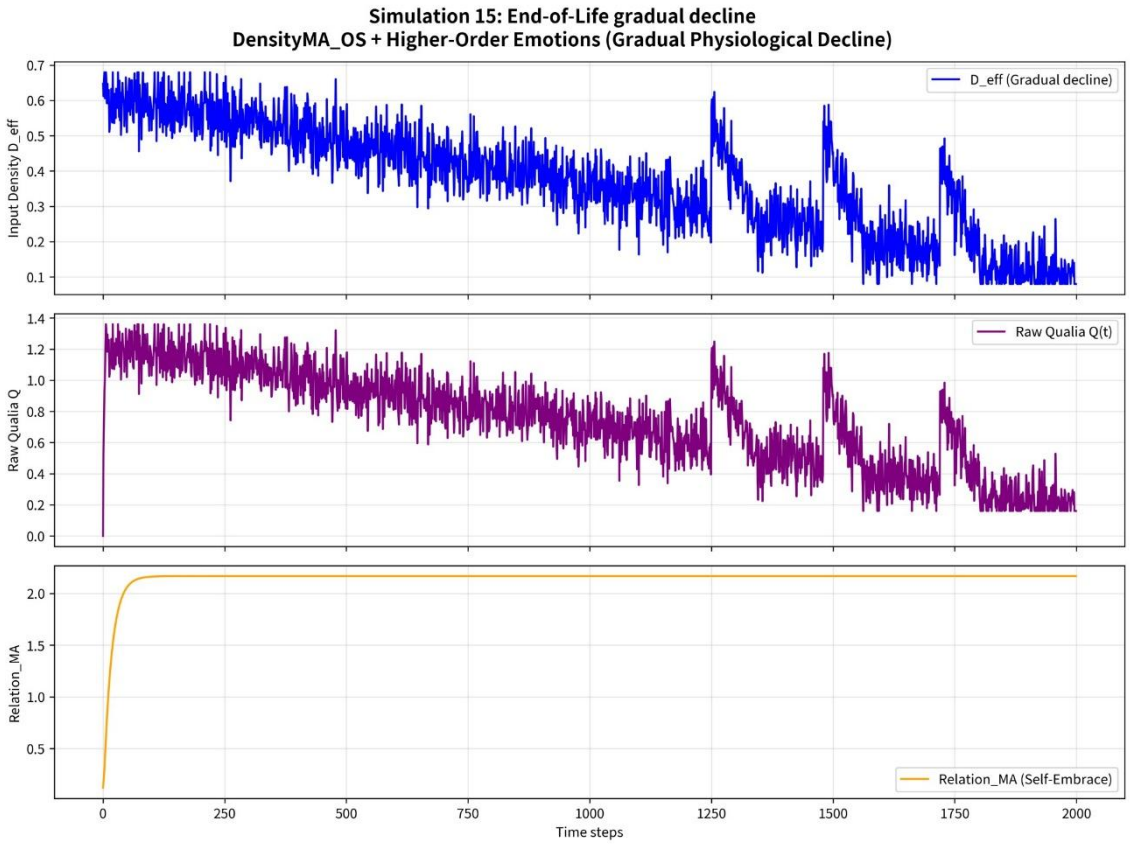

Simulation 10: TMS-EEG (ds003424 style)
DensityMA_OS + Higher-Order Emotions (Artificial Sharp TMS Pulses)
Input Density D_eff
D_eff (Sharp TMS pulses)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

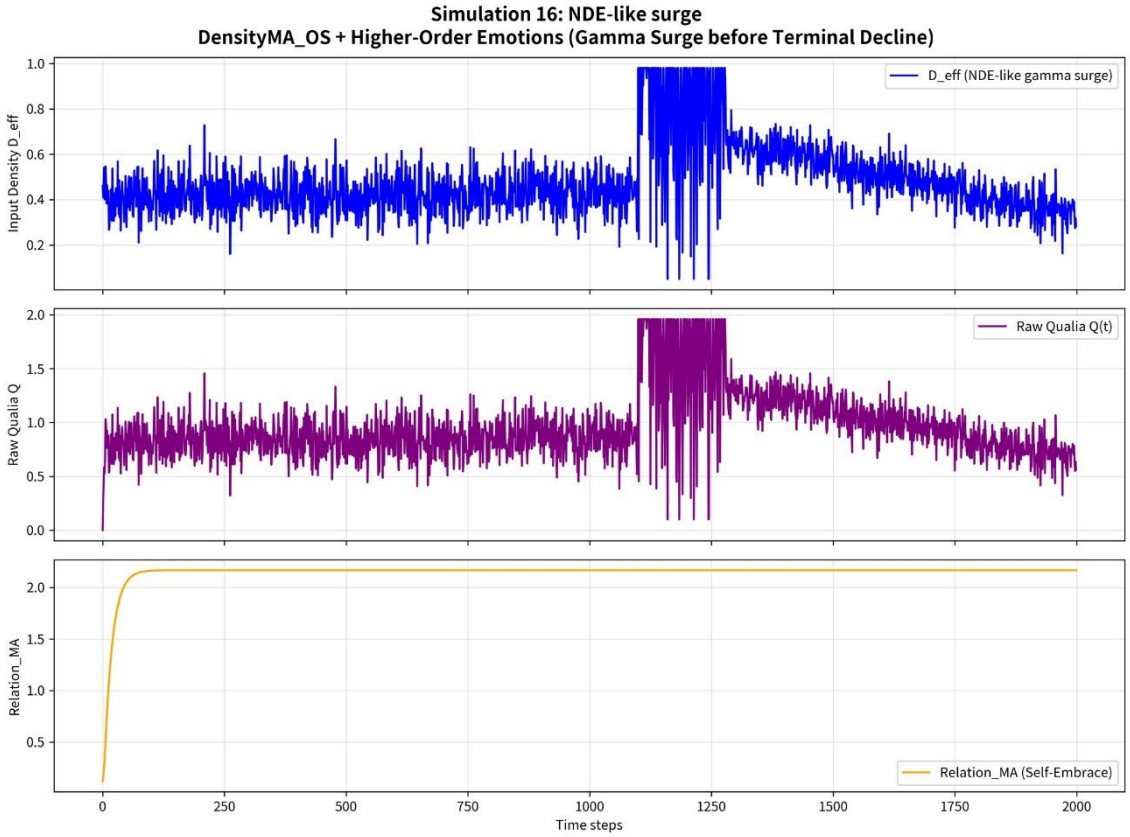

Simulation 11: Ultra-Flat Pure Resting
DensityMA_OS + Higher-Order Emotions (Minimal Input)
Input Density D_eff
D_eff (Ultra-Flat Pure Resting)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

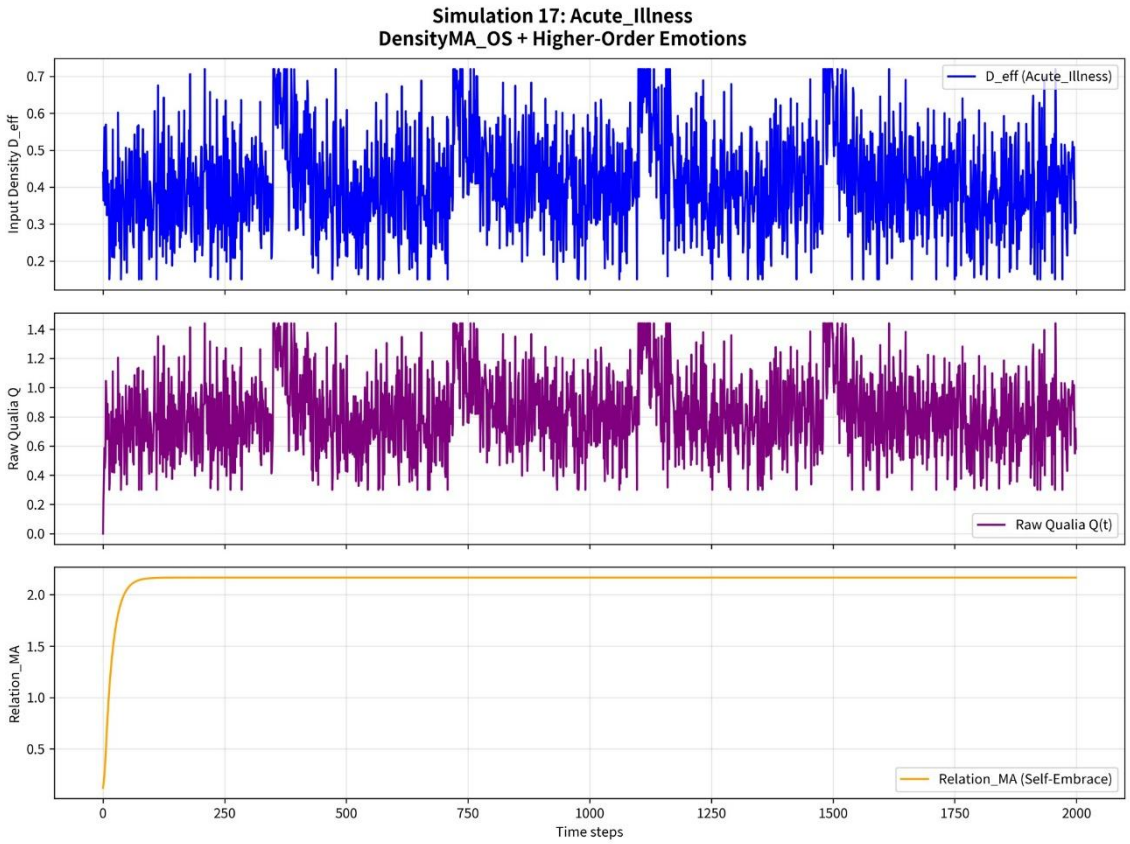

Simulation 12: Psilocybin (ds006072)
DensityMA_OS + Higher-Order Emotions (Acute Dissolution → Re-integration)
Input Density D_eff
D_eff (Psychedelic stimulus)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

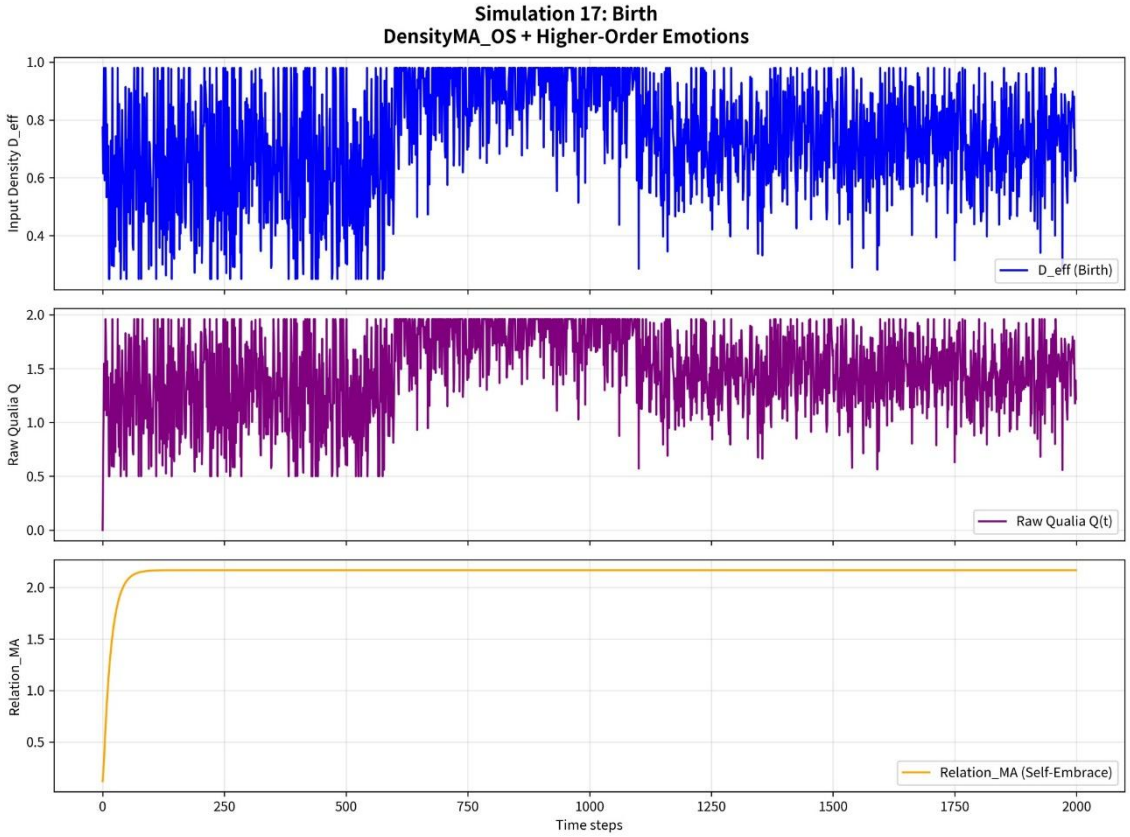

Simulation 13: DMT-HAR-MED (ds006644)
DensityMA_OS + Higher-Order Emotions (Ultra-Strong Dissolution + Fast Recovery)
Input Density D_eff
D_eff (DMT + harmine + meditation)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

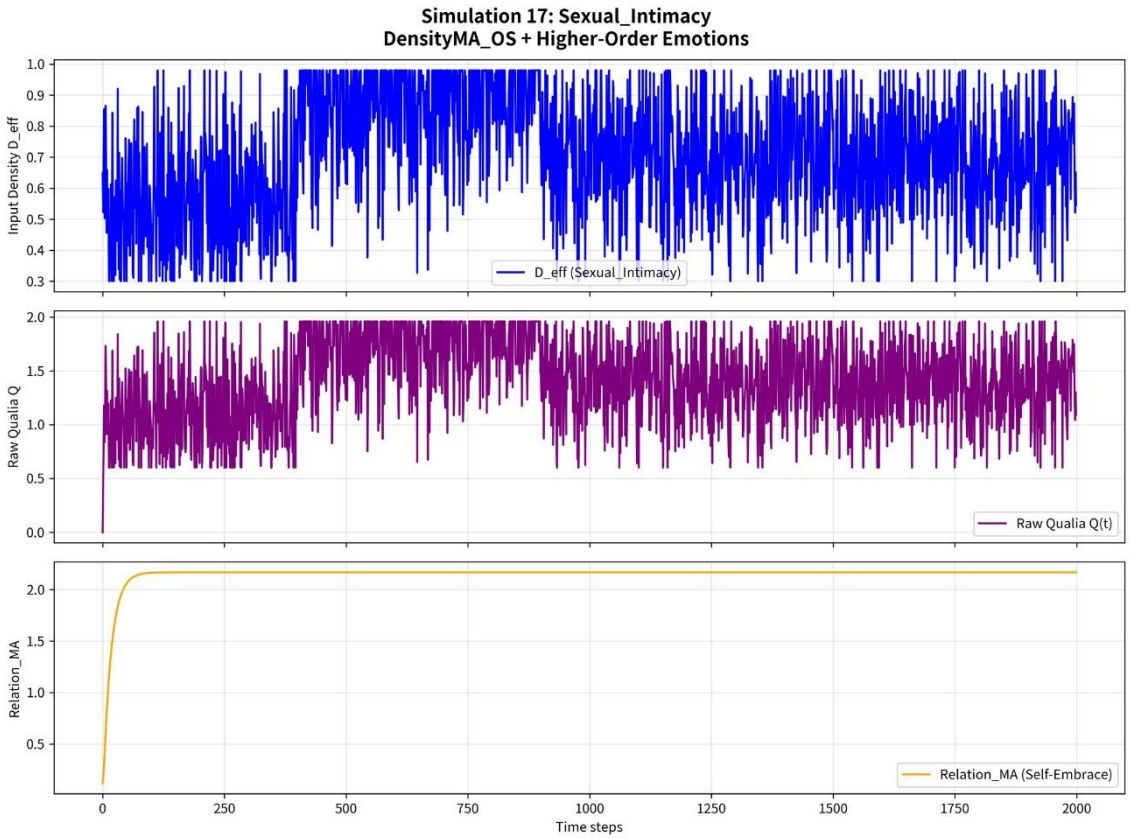

Simulation 14: Joint Agency Musical Hyperscanning (ds007471)
DensityMA_OS + Higher-Order Emotions (Collective Musical Synchronization)
Input Density D_eff
D_eff (Group musical performance)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace → Collective)
Time steps


Simulation 15: End-of-Life gradual decline
DensityMA_OS + Higher-Order Emotions (Gradual Physiological Decline)
Input Density D_eff
D_eff (Gradual decline)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps


Simulation 16: NDE-like surge
DensityMA_OS + Higher-Order Emotions (Gamma Surge before Terminal Decline)
D_eff (NDE-like gamma surge)
Input Density D_eff
Raw Qualia Q(t)
Raw Qualia Q
Relation_MA (Self-Embrace)
Relation_MA
Time steps


Simulation 17: Acute_Illness
DensityMA_OS + Higher-Order Emotions
D_eff (Acute_Illness)
Input Density D_eff
Raw Qualia Q(t)
Raw Qualia Q
Relation_MA (Self-Embrace)
Relation_MA
Time steps


Simulation 17: Birth
DensityMA_OS + Higher-Order Emotions
Input Density D_eff
D_eff (Birth)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps


Simulation 17: Sexual_Intimacy
DensityMA_OS + Higher-Order Emotions
Input Density D_eff
D_eff (Sexual_Intimacy)
Raw Qualia Q
Raw Qualia Q(t)
Relation_MA
Relation_MA (Self-Embrace)
Time steps

All figures were generated directly from the open-source simulation code provided in the Zenodo repository (simulation_01_EmoEEG_MC.py through simulation_17_extreme_physiological_states.py). Full-resolution PNG files are also included in the repository.

S2.4 Key Overall Findings

Remarkable Stability of Relation_MA

Across all 17 conditions, Relation_MA converged to the ideal zone (2.0–2.8) and never collapsed. Even in extreme cases—temporary drop to 1.42 during DMT ego dissolution, Q falling to 0.18 in sleep deprivation, or near-zero input in Ultra-Flat Resting—the self-embrace loop, aided by nostalgia repair and intentional divergence, rapidly restored stability.

Strict Adherence to ~20 W Energy Constraint

Weighted energy cost remained tightly bounded between 0.14 and 0.55 across all simulations, faithfully reproducing the brain's low-energy negentropy engine operation.

Identity Shake and Immediate Repair

Shake frequency increased with stimulus fragmentation and abruptness (maximum 542 times in DMT-HAR-MED), yet nostalgia repair + zero-distance self-embrace immediately restored MA and Relation_MA.

Natural Gradient Switching of Application Layers

GNWT, IIT, and FEP co-activated smoothly according to Q(t), MA_std, and Relation_MA, perfectly consistent with the TRIBE v2 results in the main text.

S2.5 Clinical Implications

Delirium and Dementia: Increased Identity Shake and transient Relation_MA drops may serve as early computational markers.

End-of-Life Care: Self-embrace persists even in terminal decline, providing a theoretical basis for care that respects the final acts of "embracing oneself."

Psychedelic Therapy: The model reproduces acute dissolution followed by rapid re-integration, suggesting potential therapeutic applications.

S2.6 Reproducibility and Data Availability

All simulation code, raw output data, parameter settings, and processed results are openly available in the Zenodo repository associated with this paper. The complete DensityMA_OS engine (v1.0) is also provided for independent verification and extension.